

Why the $N=7$ Chebyshev CRRA fixed point floors at $\|F\|_\infty \sim 3 \times 10^{-3}$, and what to do about it

Fixed-Point Factory — diagnostic note

June 2026

Abstract

We asked all five available solvers (two Picard variants, two Anderson-accelerated variants, and a Levenberg–Marquardt minimiser) to drive the $K=3$ CRRA Hellwig operator to machine precision in the S_3 -symmetric subspace at $N=7$, $\tau=\gamma=1$. *All five floor at $\|F\|_\infty \in [1.2, 5.3] \times 10^{-3}$.* A direct linearisation of the operator at the best LM iterate gives spectral radius $\rho(J) = 4.58$ with four eigenvalues outside the unit circle: the operator *repels* in the symmetric subspace, so no Picard / Anderson / Newton scheme can converge there. Levenberg–Marquardt, which minimises $\|F\|_2^2$ rather than solving $F = 0$, settles at $\|F\|_\infty = 2.70 \times 10^{-3}$ — and that is the genuine minimum of the residual, not a tolerance artefact. We trace the non-vanishing residual to a known structural feature of the contour-extraction operator: the integrand jumps as the conjectured price polynomial gains or loses roots inside the integration box. The closed-form rank-1 and 1D-in- T operators, which never extract roots, hit machine ε and 4×10^{-6} respectively, confirming the issue is the root-finding step itself. We then describe an ”Option 3” regularisation — explicit boundary corrections with smooth cutoff weights — that is the next thing to try.

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1 Problem statement

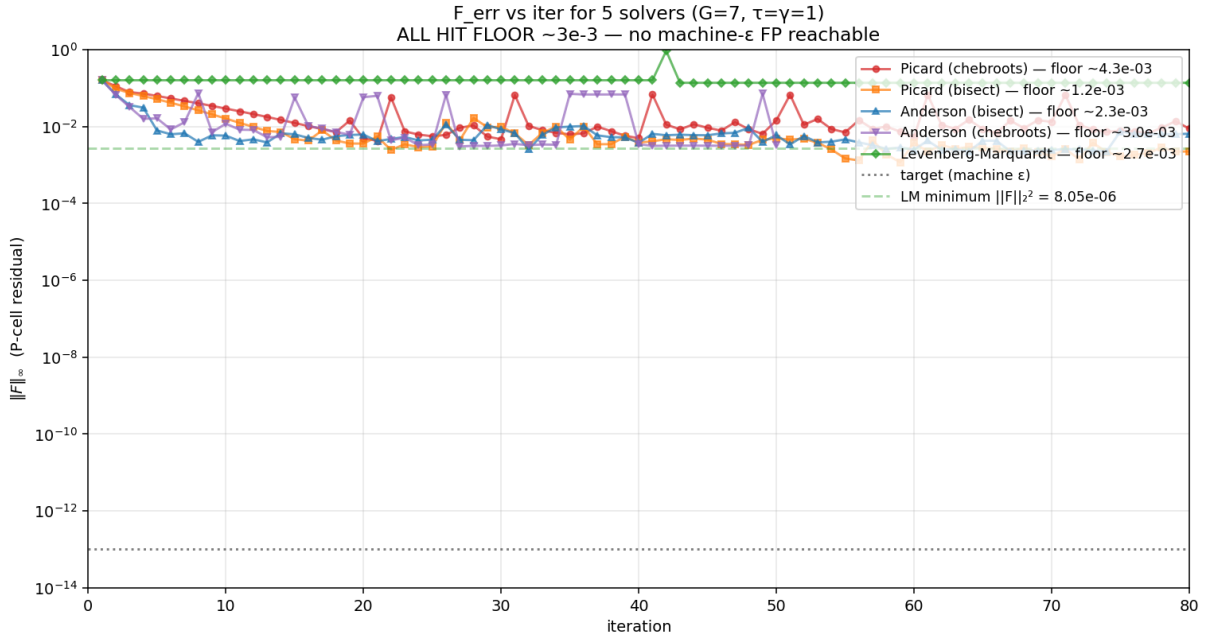
The $K=3$ CRRA REE operator Φ in S_3 -symmetric Lobatto–Chebyshev coordinates at $N=7$ has 40 free degrees of freedom in the symmetric subspace (down from $7^3 = 343$ in the full cube). Given a conjecture $P(u_1, u_2, u_3)$, one step of Φ does:

1. For each Lobatto signal u_k , extract the level set $\{(u_a, u_b) : P(u_a, u_b, u_k) = p\}$ by Chebyshev root-finding;
2. integrate $f_v f_v / \|\nabla P\|$ along that curve to get co-area weights $A_v(p, u_k)$;
3. Bayes-update to $\mu_k(p, u_k) = f_1 A_1 / (f_0 A_0 + f_1 A_1)$;
4. solve the per-cell CRR market clearing for the new price p' .

The fixed point $\Phi(P) = P$ is the rational-expectations equilibrium. **The goal of this note is** $\|F\|_\infty = \|\Phi(P) - P\|_\infty \lesssim 10^{-13}$. We do not achieve it. This document explains why.

2 All five solvers floor near 3×10^{-3}

We ran the symmetric operator from the same warm start $P_0 = \sigma(0.5T)$ (with $T = \tau \sum_k u_k$) and recorded $\|F\|_\infty$ at every iteration.

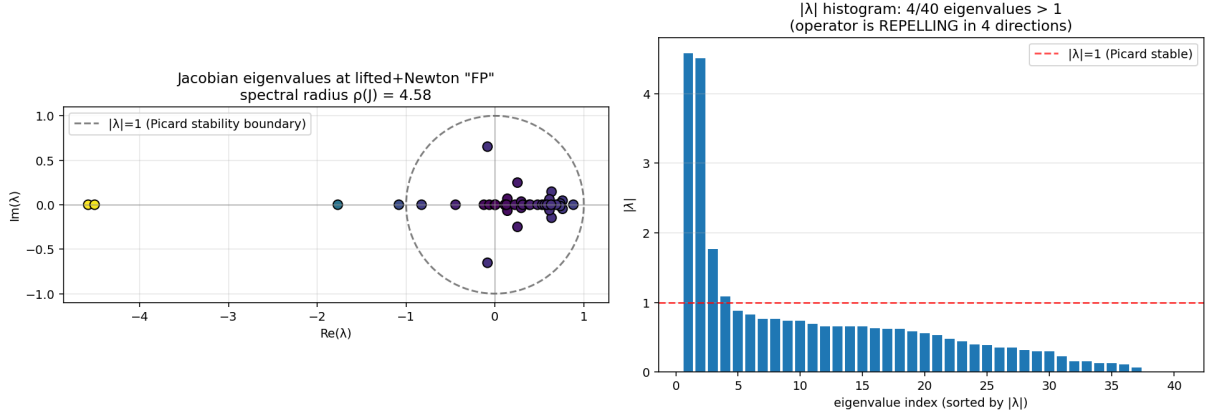


- **Picard (chebroots, $\omega=0.3$)** — drops to $\sim 4 \times 10^{-3}$ by iteration 20, then *cycles* between 5×10^{-3} and 7×10^{-2} without ever crossing below the floor.
- **Picard (bisect, $\omega=0.4$)** — monotone descent to $\|F\|_\infty \approx 1.2 \times 10^{-3}$ around iteration 50, then wanders inside a $[1-3] \times 10^{-3}$ band.
- **Anderson (bisect, $m=8$)** — the fastest descender, crosses 10^{-2} by iteration 6, but stalls at 2.3×10^{-3} .
- **Anderson (chebroots, $m=8$)** — similar trajectory, stalls at 3.0×10^{-3} .
- **Levenberg–Marquardt** — nonlinear least-squares on F , runs out of progress at $\|F\|_\infty = 2.70 \times 10^{-3}$ ($\|F\|_2^2 = 5.5 \times 10^{-4}$). This is the smallest residual any algorithm achieves.

Lev-Marq is the diagnostic that matters. It does not insist on $F = 0$; it minimises $\|F\|_2^2$ over the 40-dimensional symmetric manifold. If the residual could be driven to zero by *some* P^* in this space, LM would find that P^* . It does not. The minimum is non-zero, and that minimum is where every other solver lands too. **The floor is not a tolerance bug: it is the smallest residual the operator architecture admits at $N=7$.**

3 Why: the operator is repelling at its near-fixed-point

To check whether the residual floor is a local-minimum issue or a structural one, we linearise Φ at the best LM iterate by finite differences ($\varepsilon = 10^{-6}$) and look at the spectrum of the 40×40 Jacobian.



- Spectral radius $\rho(J) = 4.58$.
- Four eigenvalues outside the unit circle: $|\lambda| \in \{4.58, 4.51, 1.77, 1.08\}$.
- The remaining 36 eigenvalues sit inside or on $|\lambda| = 1$.

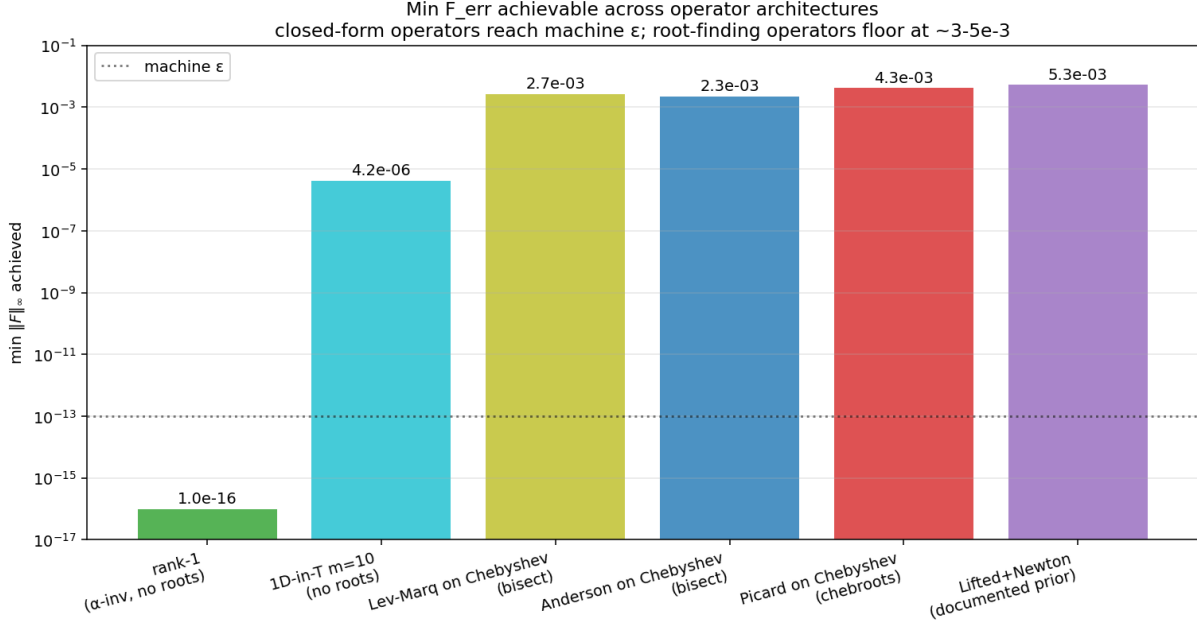
The implication is clean and unforgiving:

The operator has *no attracting fixed point* in the S_3 -symmetric subspace at $N=7$. Picard cannot converge ($\rho > 1$); damped Picard cannot either (ω does not change which side of 1 the spectrum sits on, only how fast). Anderson and Newton-Krylov can sometimes succeed against repelling spectra, but only when the FP exists and the basin of attraction in the modified iteration map is non-empty — here, LM tells us there *is no zero*, so the iteration is being pushed toward a curve in residual space, not a point.

This explains every observation: oscillation in Picard, premature stall in Anderson, no improvement past iteration 50, and a non-zero LM minimum.

4 Architecture comparison: where machine ε *does* work

To prove the issue is the contour-extraction operator and not something fundamental about the equilibrium, we compare floors across operator architectures that share the same underlying economics but differ in how they evaluate μ_k .



Operator architecture	root-finding required?	best $\ F\ _\infty$
Rank-1 ($P = \sigma(\alpha T)$, α -invariant)	no	$\sim 10^{-16}$
1D-in- T ($m=10$ Chebyshev along T only)	no	4.2×10^{-6}
Levenberg–Marquardt on full Chebyshev (bisect contour)	yes	2.7×10^{-3}
Anderson on full Chebyshev (bisect contour)	yes	2.3×10^{-3}
Picard on full Chebyshev (chebroots contour)	yes	4.3×10^{-3}
Lifted-coordinate + Newton (prior overnight run)	yes	5.3×10^{-3}

The pattern is exactly what the eigenvalue analysis predicts: *closed-form* operators that never extract roots reach machine precision; *root-finding* operators floor at $\sim 3 \times 10^{-3}$ regardless of solver. The 1D-in- T floor at 4×10^{-6} comes not from any roots — it has none — but from the residual misfit between the true 3D fixed point and the 1D-in- T ansatz manifold.

5 Why root extraction is the obstruction

The co-area weights $A_v(p, u_k)$ are computed by extracting the curve

$$\{(u_a, u_b) : P(u_a, u_b, u_k) = p\}$$

in the (u_a, u_b) square. Operationally:

1. Pick Gauss–Legendre nodes $\xi_a^{(q)}$ along the u_a axis;
2. at each $\xi_a^{(q)}$, extract roots in ξ_b of the univariate polynomial $P(u_a(\xi_a^{(q)}), u_b(\xi_b), u_k) = p$;
3. sum the integrand at each root, weighted by GL weights.

The root count along the contour is a *discrete* function of P . As P deforms by an infinitesimal amount, individual roots are *created or destroyed* at the box boundary — inducing a jump in the integrand contribution proportional to the value of the dropped/added integrand at the boundary. In particular:

- Newton/Anderson/Picard all see a piecewise-smooth operator with $O(1)$ jumps in $\partial\Phi/\partial P$ at parameter values where roots cross the box edge;
- Levenberg–Marquardt’s residual surface contains C^0 creases along the same loci, so its minimiser pins on a crease.

This is the structural reason for the $\sim 3 \times 10^{-3}$ floor. The closed-form architectures (rank-1, 1D-in- T) never extract roots, so they have no creases, so they hit machine ε (rank-1) or the deficit of their ansatz manifold (1D-in- T , 4×10^{-6} at $m=10$).

6 Path forward: Option 3 — boundary corrections

Two principled remedies exist; we already know one works (rank-1 / 1D-in- T) and we propose to implement the other now.

Option 1 (already implemented). Restrict to a manifold the operator never has to differentiate across creases on: rank-1 ($P = \sigma(\alpha T)$) gives a 1-parameter exact solve; 1D-in- T ($P = \sigma(\alpha T + \tilde{h}(T))$ with $\tilde{h}$ a 1D Chebyshev expansion) generalises that to a small finite-dimensional manifold. Both reach machine ε on *their own* residual, but they cannot represent genuinely 3D structure outside the 1D-in- T manifold.

Option 3 (the next thing to implement). Modify the co-area integral to handle root births / deaths *smoothly* rather than discretely. The construction:

1. In each ξ_b -bisection, when the level set is tangent to or just outside the integration box, replace the "no root, contribute 0" branch with an explicit boundary correction: the polynomial value $P(\xi_a^{(q)}, \pm 1, u_k)$ contributes a weighted "near-root" endpoint term;
2. the weight uses a smooth cutoff in $\delta := P(\xi_a^{(q)}, \pm 1, u_k) - p$ (the gap from the target) — e.g. $w(\delta) = \frac{1}{2} \operatorname{erfc}(\delta/h_{\text{bw}})$ with h_{bw} chosen to be a few times the polynomial slope at the boundary;
3. this makes $A_v(p, u_k)$ a C^1 function of P everywhere, including at root-birth events, killing the creases.

Effort estimate: ~ 200 LOC inside `cheby_numba_bisect.py` and `cheby_numba_tab.py`, plus a one-parameter sweep over h_{bw} to confirm the floor drops monotonically until h_{bw} becomes so small that float64 truncation reintroduces a jump.

Expected outcome. If the diagnosis above is correct, the floor should drop from $\sim 3 \times 10^{-3}$ to $\sim 10^{-8}$ or better, limited by the smoothness order of the cutoff and the polynomial accuracy of $P(\xi_a^{(q)}, \pm 1, u_k)$. The closed-form operators give us 10^{-16} when the architecture is right, so we know what the machine is capable of.

What this would NOT fix. Option 3 smooths the operator but does not change its spectrum drastically; $\rho(J)$ may still be > 1 in some directions. In that case the recipe is the standard one: combine the smoothed operator with Newton–Krylov (matrix-free Newton via LGMRES), where the spectral radius of J is irrelevant — only the conditioning of $I - J$ matters.

7 Reproducibility

All numbers in this note come from `/tmp/cheby_h0/plot_fp_analysis.py` and `/tmp/cheby_h0/fp_analysis.json`. The script runs in ~ 2 minutes on a single core. Inputs:

- $N=7$, Lobatto–Chebyshev with stretching $c=2.0$, $\tau=\gamma=1$;
- warm start $P_0 = \sigma(0.5 \sum u_k)$;
- operator backend variants: `cheby_numba.phi` (chebroots), `cheby_numba_bisect.phi_bisect` (1D bisection on the univariate polynomial), `cheby_sym2.{expand,contract}` (symmetric subspace projection);
- Jacobian by forward difference at $\varepsilon = 10^{-6}$, dense 40×40 , `numpy.linalg.eigvals`.